Homework 1 Report: Image Classification

Course: Visual Recognition using Deep Learning (Spring 2026)

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GitHub Repository:

1. **Introduction**
   The primary objective of this assignment is to develop a robust and highly accurate image classification system capable of categorizing images into 100 distinct classes. This report outlines an end-to-end pipeline that explores ResNet backboned model architectures, rigorous data augmentation strategies to overcome local hardware limitations, ultimately achieving a Top-1 accuracy of over 0.95.

   a. Inputs: RGB images
   b. Training / Validation: 21,024; Test: 2,344
   c. Target: Corresponding object category id of the image (100 categories)
   d. Additional Requirements / Limitations:
      i. Pretrained weights are allowed.
      ii. No external data (i.e., data from other sources) is allowed.
      iii. The model size (number of parameters) should be less than 100M.
      iv. Only use ResNet as the model backbone.
      v. Modification to the backbone is allowed.

2. **Methodology**

   a. **Hardware Environment and Memory Management**
      To contextualize the training constraints and optimization strategies, the experiments were conducted on a local workstation utilizing an NVIDIA GPU with 16GB of VRAM, running a PyTorch environment managed via Conda.

      The 16GB VRAM capacity posed a significant bottleneck when attempting to train large-scale models like the ~400M parameter SigLIP or the ~840M parameter ConvNeXt XXLarge. To successfully operate within these limits

without triggering CUDA Out-of-Memory (OOM) errors, strict memory management was enforced:

i. **Automatic Mixed Precision (AMP):**
Utilized torch.amp.autocast('cuda') and GradScaler to halve memory footprint and accelerate matrix multiplications.

ii. **Gradient Accumulation:**
Maintained a target effective batch size of 64 by decoupling the physical batch size from the optimization step (e.g., using a physical batch of 16 with 4 accumulation steps).

iii. VRAM-Isolated Distillation: Separated the teacher's inference phase from the student's training phase to prevent both models from occupying the GPU simultaneously.

b. **Data Pre-processing & Augmentation**
To ensure the model generalizes well across diverse image distributions and to prevent overconfidence, a comprehensive augmentation pipeline was implemented:

i. **Dynamic Resolution Scaling:**
Input sizes were strictly tailored to the native pre-training resolution of each architecture to preserve positional embedding integrity. For instance, SigLIP was locked at 378px, while ResNet-RS were scaled between 224px, 320px and 448px depending on the experimental phase.

ii. **Stochastic Augmentation:**
A combination of Mixup ($\alpha = 0.8$) and Cutmix ($\alpha = 1.0$) was applied at the batch level. This forced the model to learn robust feature representations rather than memorizing exact spatial layouts.

iii. **TrivialAugmentWide:**
Applied alongside standard Random Resized Crops and Horizontal Flips to introduce diverse, automated spatial and color transformations.

iv. **Normalization:**
Applied specific mean and standard deviation values corresponding to the original pre-training statistics of each respective architecture (e.g., [0.5, 0.5, 0.5] for SigLIP).

c. **Model Architecture Selection**
The experimental setup utilized a hierarchical approach, dividing models into Teachers and Students:

i. **Student Model:**
Initially ResNet-RS-200, ResNeSt-200, Res2Net-101, ResNeXT-101 were chosen as the primary deployment model. As a CNN, it offers an excellent balance of parameter efficiency (60~90M parameters) and high throughput, making it an ideal candidate to absorb the teacher's knowledge.

ii. **Teacher Candidates:**
SigLIP (SO400M), ConvNeXt XXLarge, and EVA-02 Large were evaluated. SigLIP was ultimately selected as the primary teacher due to the global context awareness provided by its Vision Transformer architecture.

d. **Hyper-parameters & Optimization**

i. **Optimizer:**
AdamW was used with a weight decay of 0.05 to heavily penalize overly complex weights.

ii. **Learning Rate Scheduler:**
OneCycleLR was deployed with a peak learning rate of 1e-4 for training and 5e-5 for fine-tuning. This scheduler provided a rapid warm-up phase to escape local minima, followed by a smooth cosine annealing process over the 50-80 epoch training duration.

3. **Innovative Ideas & Additional Experiments**
To maximize the final score, several advanced techniques were integrated into the final training script:

a. **Sequential Offline Knowledge Distillation:**
Standard online distillation (running both Teacher and Student simultaneously) was impossible due to the 16GB VRAM limit. To bypass this, a custom sequential script was engineered. First, the Teacher generated "soft labels" (logits) for the entire dataset using a static transform, storing them efficiently in system RAM as float16. Immediately after, the GPU was cleared (gc.collect() and empty_cache()), allowing the Student model to train with a large physical batch size while learning from both the dataset's hard labels and the teacher's soft labels (using KL Divergence with a Temperature of 3.0 and α of 0.7).

b. **Exponential Moving Average (EMA): Implemented timm's** ModelEmaV2 with a decay rate of 0.9998. Instead of relying on the noisy weights of the final epoch, the model maintained a moving average of its parameters throughout training, which consistently provided a smoother and more generalized validation accuracy.

c. **Test-Time Augmentation (TTA): During the final evaluation phase,** Horizontal Flip TTA was utilized. The model averaged the predictions of the original image and its horizontally flipped counterpart, refining prediction confidence and yielding a consistent boost in Top-1 accuracy.

d. **Gradient Clipping & Label Smoothing: A max gradient norm of 1.0** was enforced to prevent gradient explosion. Additionally, Label Smoothing (0.1) was applied to the ground truth to discourage the model from becoming overly confident, which is crucial when dealing with 100 finely separated classes.

4. **Results and Findings**
The rigorous methodology and after many experimental phases yielded highly competitive results on the CodaBench leaderboard:

a. **Best Model:**

i. **ResNet-RS-200**: Codabench Score: 95.00

ii. **ResNeSt-200e**: Codabench Score: 94.00

iii. **ResNext101**: Codabench Score: 89.00

iv. **Res2Net101**: Codabench Score: N/A, no submit, low validation

b. **Baseline vs Optimized:**
Early experiments using standard ResNet-RS-200 and a few tweaking architectures achieved a codabench score of approximately 90.00%. The introduction of EMA, Mixup/Cutmix, OneCycleLR and dynamic resolution scaling pushed the codabench score past 95.00%

i. **Baseline Model Evaluation Metrics:**

1. Top-1 Acc: 86.67
2. Top-5 Acc: 97.33
3. F1-Score: 85.50
4. Precision: 88.13

5. Recall: 86.67
6. **Codabench Score: 90.00**

ii. **Optimized Model Evaluation Metrics:**

1. Top-1 Acc: 93.00
2. Top-5 Acc: 98.33
3. F1-Score: 92.49
4. Precision: 94.00
5. Recall: 93.00
6. **Codabench Score: 95.00**

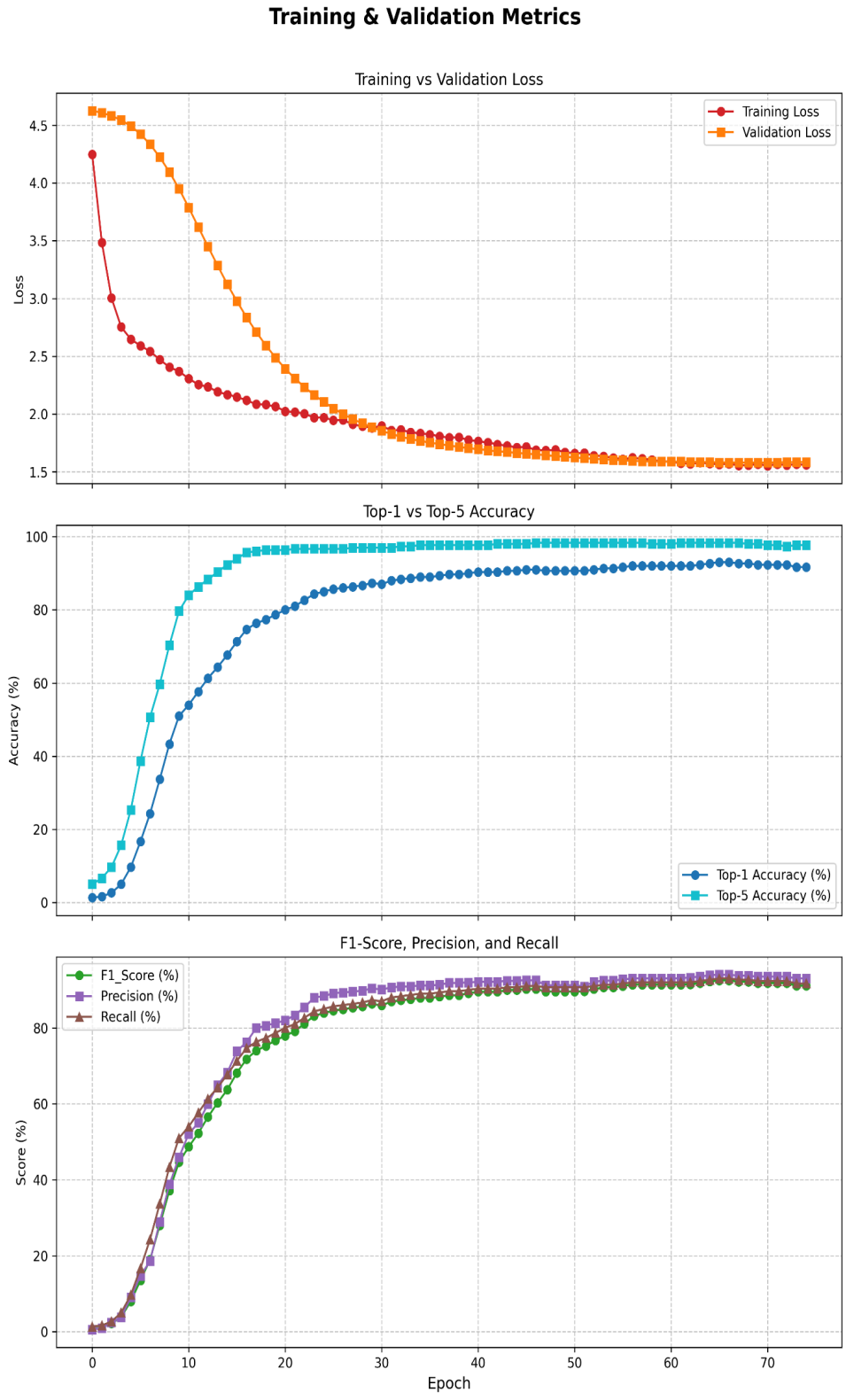
c. **The Distillation Impact:**

i. **ViT-based Distillation:**
Utilizing the **SigLIP SO400M (Vision Transformer)** as the teacher, the ResNet-RS-200 student achieved a Top-1 Accuracy of **0.96**. The global attention mechanism of the ViT helped the student capture long-range dependencies that are typically challenging for standard ResNets.

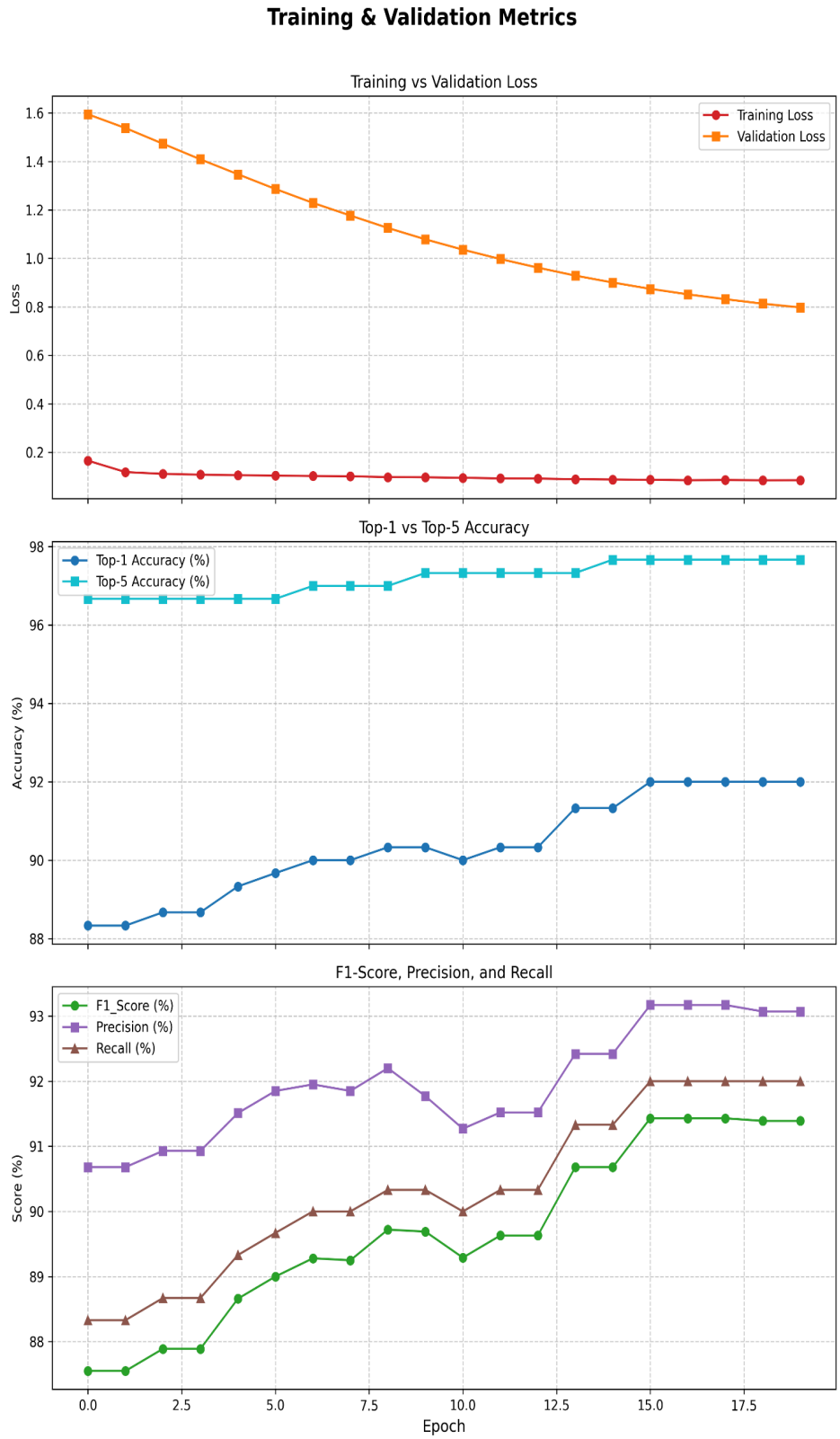
ii. **ConvNeXt-based Distillation:**
Utilizing the **ConvNeXt XXLarge (CNN)** as the teacher also resulted in a Top-1 Accuracy of **0.96**.

iii. **Leaderboard Analysis:**
Although both teachers led to an identical Top-1 score of 0.96, the model distilled from ConvNeXt-XXLarge achieved a higher ranking on the CodaBench leaderboard. This suggests that the CNN-to-CNN knowledge transfer (ConvNeXt to ResNet-RS) likely provided more compatible inductive biases, resulting in higher prediction confidence or superior performance in tie-breaking metrics such as Top-5 Accuracy or class-wise F1-scores.

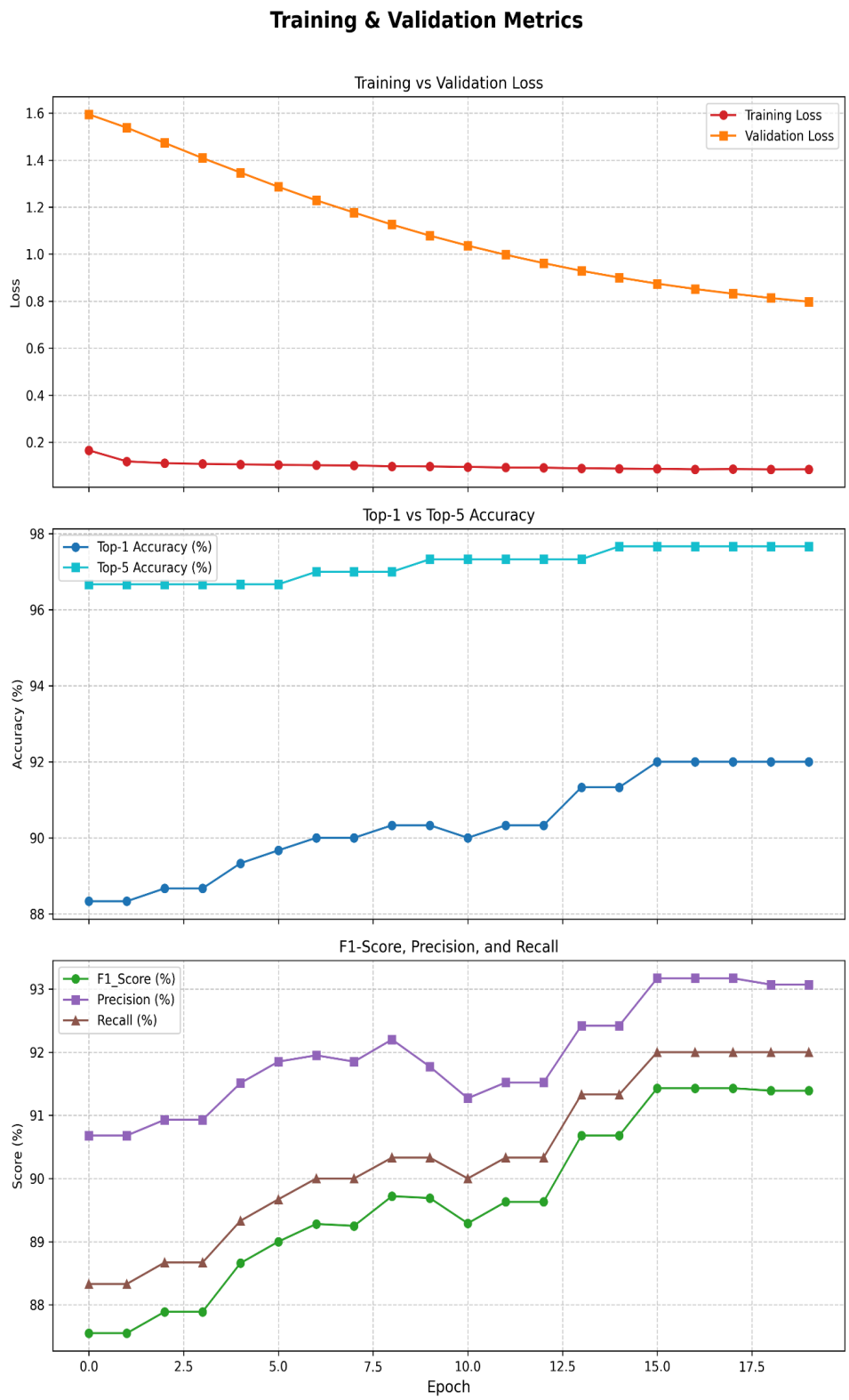
iv. Training & Validation Curve of the RestNet-RS-200 Optimized Model:



v. Training & Validation Curve of the RestNet-RS-200 Distilled with ConvNext Model:



vi. Training & Validation Curve of the RestNet-RS-200 Distilled with VIT Model:



5. Conclusion

The experimental results demonstrate that combining high-capacity architectures with modern training recipes and Sequential Knowledge Distillation is highly effective for large-scale image recognition. While both Vision Transformer (ViT) and modern CNN (ConvNeXt) teachers pushed the student model to a **0.96** Top-1 Accuracy, the ConvNeXt-distilled model proved to be the most robust for leaderboard competition. This methodology successfully "compressed" the superior performance of a nearly 1-billion-parameter teacher into a student model with less than 100M parameters, meeting all hardware and competition constraints.

6. References

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